From Leiden to Pleasure Island: The Constant Potts Model for Community Detection as a Hedonic Game
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November 11, 2025

Motivation: Clusters or Community

- Unsupervised Machine Learning Technique
- Networks
- Non-Overlapping Partitions
- Competing Objectives
  - Maximize Friends
  - Minimize Strangers
- Frustrated Choice (Cannot Max/Min both)

Background: Robustness

- A node is **robust** if its current community is unambiguously preferred over all other communities.
- For a given partition, the robustness will be the fraction of nodes that there is no better move for.

Background: Familiarity Index

$$F_i(C_A, C_B) = \frac{\Delta d}{\Delta d + \Delta \hat{d}}$$

- $F > 1$ or $F < 0$ you should make that change
- $0 < F < 1$ are frustrated choices

Background: Constant Potts Model

Comes from an analog of the Hamiltonian for the magnetic spin in statistical mechanics.

$$v_{ij} = (A_{ij} - \gamma)\, \delta_{v_i, v_j \in C_k}$$

Can vary γ:

- $\gamma = 0$ ignores strangers
- $\gamma = 1$ ignores friends
- Frustrated nodes
  - $\gamma < F_i$ go to more friends
  - $\gamma > F_i$ go to less strangers

Methodology: Game Theory

- Agents (nodes here) act selfishly
- Cooperation happens (stable communities emerge)
- Nash Equilibrium happens
- Equilibrium is beneficial for collective

Methodology: Hedonism

- Hedonic Potential created
- Utility Functions can be computed in polynomial time
- Three scales:
  1. Node

$$\varphi_i^{\gamma}(C_k) = \sum_{j \in C_k} v_{ij}$$

  2. Community

$$\phi^{\gamma}(C_k) = \frac{1}{2} \sum_{i \in C_k} \sum_{j \in C_k} v_{ij}$$

  3. Partition

$$\Phi^{\gamma}(\pi) = \sum_{k=1}^{K} \phi^{\gamma}(C_k) = (1-\gamma) \sum_{k=1}^{K} m_k - \gamma \sum_{k=1}^{K} \left(\binom{n_k}{2} - m_k \right)$$

Methodology: Leiden Algorithm

1. Local Moving of Nodes: Uses a random queue and only looks at moving neighbors after evaluating a node
2. Refinement or Partition: Look into a community and makes random but improving choices (not greedy!)
3. Aggregation of Network: Take the refinements and reorganize them into communities

Methodology: Game Equilibrium

- Stability (or Equilibrium)
  - Internally Stable Communities: No node inside a community has incentive to leave
  - Externally Stable Communities: No node outside community has incentive to join
  - Locally Stable Node: Nodes do not have incentive to deviate
- Solution is partition with all nodes locally stable

Results and Findings: Overview

- γ is rational
- Guaranteed convergence
- pseudo-polynomial time

Why?

- Φ is bounded within $[-n^2, n^2]$ and increasing
- Since $\gamma = \frac{b}{c}$, each beneficial node increases potential by $\frac{1}{c}$.
- Convergence happens in $O(cn^2)$

Results and Findings: Pseudo Code

Algorithm 1: **Better-Response Dynamics for Finding a Hedonic Game Equilibrium.**

Input : $G = (V, E)$; γ; Initial partition membership $\sigma : V \rightarrow \{1, \ldots, K\}$.
Output: Equilibrium partition σ.

changed ← true ; // Ensures the main loop runs at least once
while *changed* **do**
    changed ← false
    Initialize queue $Q \leftarrow V$
    while *Q is not empty* **do**
        Remove a node i from Q ; // Select a candidate node to evaluate
        if *Node-level best-response* **then**
            $\mathcal{K}^\star \leftarrow \arg\max_{k \in \{1,\ldots,K\}} \varphi_i^\gamma(C_k)$; // Best communities for node i
        else
            $\mathcal{K}^\star \leftarrow \{k \in \{1, \ldots, K\}$ such that $\varphi_i^\gamma(C_k) > \varphi_i^\gamma(C_{\sigma_i})\}$; // Better communities for node i
        end
        Let κ be an element from $\mathcal{K}^\star$; // Select one target community (handling ties)
        if $\varphi_i^\gamma(C_\kappa) > \varphi_i^\gamma(C_{\sigma_i})$ **then**
            $\sigma_i \leftarrow \kappa$
            changed ← true
            $Q \leftarrow Q \cup \{j \in V \mid A_{ij} = 1$ and $j \notin C_\kappa\}$; // Leiden's local move queue management [6]
        end
    end
end
return σ ; // Return partition as solution according to Definition 3

Results and Findings: Evaluation

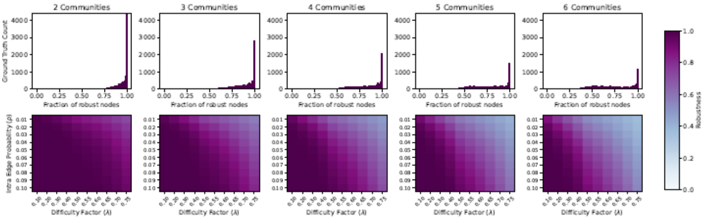

Figure 7: The top row displays histograms illustrating the distribution of the fraction of robust nodes across various parameter settings. The bottom row presents heatmaps, with each cell showing the average fraction of robust nodes for a given connection probability (p) and difficulty factor (λ). This figure demonstrates how the robustness (defined in Eq. (1)) of the ground-truth partition changes with the difficulty factor. Each heatmap corresponds to a different number of communities ($K \in \{2, 3, 4, 5, 6\}$), keeping $n = 1020$ fixed and $N = 1020/K$. Darker colors indicate a higher fraction of fully-robust nodes.

Critical Reflection

- New/Novel Results
- Feel of Meta-Heuristics

Citation